

Project Name:
**Smart Pharmacy Inventory Optimization System
(SPIOS)**

Domain:
Pharmacy Inventory Management

Role:
Business Analyst

Responsibilities:

- * Requirements Elicitation
- * BRD & SRS Preparation
- * BPMN Modeling
- * Use Case Modeling
- * Wireframing

Tools Used:

- * Microsoft Word
- * Draw.io
- * Figma
- * Visio

Section 1 → BRD

- 1. Business Problem**
- 2. Business Objectives**
- 3. Stakeholders**
- 4. Scope**
- 5. Business Rule**

- **Business Problem:**

The pharmacy is experiencing inventory management challenges that lead to financial losses and operational inefficiencies.

Current issues include:

1. Overstocking of slow-moving products, resulting in increased holding costs and higher risk of product expiration.
2. Stock shortages of fast-moving medications, leading to missed sales opportunities and reduced customer satisfaction.
3. Financial losses due to expired medications.
4. Lack of accurate forecasting when placing purchase orders, causing pharmacists to rely on manual judgment.

The pharmacy requires a system that analyzes historical sales data and inventory movements to provide intelligent stock recommendations, reduce expiries, and improve product availability.

- **Business Objectives:**

BO-001

Reduce inventory losses caused by expired medications by providing early expiry alerts, near-expiry reporting, and expiry monitoring capabilities.

BO-002

Improve product availability by ensuring critical medications remain in stock.

BO-003

Reduce stockout incidents by at least 40% through automated reorder recommendations.

BO-004

Optimize inventory investment by minimizing overstocking of slow-moving products.

BO-005

Improve inventory planning accuracy by replacing manual reorder decisions with data-driven recommendations.

BO-006

Increase operational efficiency by providing pharmacists with automated daily replenishment reports.

BO-007

Provide management with real-time visibility into inventory performance through dashboards and analytical reports.

BO-008

Support informed inventory decision-making by analyzing product movement trends and sales performance.

BO-009

Improve inventory control by maintaining accurate batch-level tracking for quantities and expiry dates.

BO-010

Increase accountability and compliance by recording inventory-related actions and approval activities in the audit log.

BO-011

Enable proactive management of inventory risks by identifying near-expiry and expired products before they impact business performance.

BO-012

Improve branch-level inventory governance by implementing an approval workflow for modified reorder quantities.

- **Success Metrics**

SM-001

Reduce expired inventory value by at least 30% within six months of system implementation.

SM-002

Reduce stockout occurrences by at least 40% within six months of system implementation.

SM-003

Improve reorder decision accuracy by utilizing automated recommendation calculations.

SM-004

Reduce manual inventory review effort through automated replenishment and expiry reports.

SM-005

Provide 100% visibility of products approaching expiry through dashboard monitoring and reporting.

- **Stakeholders:**

Primary Stakeholders

- 1- Pharmacist
- 2- Branch Manager

Secondary Stakeholders

- 1- Pharmacy Owner
 - 2- External POS System
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- **Scope:**

*** In Scope**

1. Product Management
2. Batch Management
3. Purchase Invoice Management
4. Inventory Tracking
5. Expiry Tracking
6. Expiry Alerts
7. Near Expiry Reporting
8. Reorder Recommendations
9. Daily Replenishment Report
10. Inventory Dashboard
11. Sales Dashboard
12. Fast-Moving Product Analysis
13. Slow-Moving Product Analysis
14. Reorder Approval Workflow
15. Audit Logging

Out of Scope

1. HR Management
 2. Payroll Management
 3. Accounting Management
 4. Supplier Management
 5. Warehouse Management System
 6. Procurement System
-

- **Business Rules**

BR-001

Every product must have a unique Product ID.

BR-002

A product cannot be created without mandatory product information.

BR-003

Each batch must belong to an existing product.

BR-004

Each product may have multiple batches, but duplicate batch numbers for the same product shall update the existing batch quantity instead of creating a new batch.

BR-005

Expired batches cannot be added to the system.

BR-006

Expiry alerts shall be generated when a batch has 90 days or less before expiry.

BR-007

The system shall generate reorder recommendations during the daily replenishment process when stock $\leq$ threshold.

BR-008

The system shall use the previous 6 months sales history when calculating reorder recommendations.

BR-009

The system shall consider the current stock quantity when calculating reorder recommendations.

BR-010

The system shall consider product expiry risk when calculating reorder recommendations.

BR-011

The system shall consider product movement classification when calculating reorder recommendations.

BR-012

The system shall generate reorder recommendations that cover 30 days of expected demand.

BR-013

If the calculated quantity is less than or equal to zero, the system shall not generate a reorder recommendation.

BR-014

The system shall add every product reorder recommendation to The Daily Replenishment Report.

BR-015

The system shall generate the Daily Replenishment Report start of each business day.

BR-016

Pharmacists may modify recommended reorder quantities before approval.

BR-017

A justification is mandatory whenever a pharmacist modifies a recommended reorder quantity.

BR-018

Modified Daily Replenishment Report quantities require Branch Manager approval.

BR-019

The Branch Manager may:

- * Approve pharmacist quantity
- * Approve original recommendation
- * Enter custom quantity
- * Reject

BR-020

All reorder actions shall be recorded in the audit log.

BR-021

Products shall remain in the Daily Replenishment Report until action is completed.

BR-022

The system shall classify products based on average 3 months sales:

Fast > Configurable value

Medium >= Configurable value and < Configurable value

Slow < Configurable value

BR-023

Products threshold shall be configurable by the Pharmacist and it must be greater than zero.

BR-024

Recommended Quantity shall be measured by Base Quantity and adjusted by Expiry Risk.

Base Quantity =

(Expected 30-Day Demand – Current Stock)

- IN High Risk Of Expiry

Recommended Quantity = Base Quantity x 70%

- IN Medium Risk Of Expiry

Recommended Quantity = Base Quantity x 90%

- IN Low Risk Of Expiry

Recommended Quantity = Base Quantity

BR-25

A batch shall automatically be marked as Near Expiry when it has 90 days or less before expiry.

BR-026

A batch shall automatically be marked as Expired at the start of the day following its expiry date.

BR-027

Expired batches shall not be included in reorder calculations.

BR-028

Disposed quantities shall be deducted from inventory automatically.

BR-029

Disposed batches shall remain in the system for audit and reporting purposes.

Section 2 → SRS

1. FRs
2. NFRs
3. User Stories
4. Acceptance Criteria
5. Data Dictionary
6. Use Cases
7. BPMN Diagrams
8. Use Case Diagram
9. Wireframes

- **Functional Requirements:**

Product & purchase invoice module

FR-001

The system shall allow authorized users to create new products.

FR-002

The system shall allow pharmacists to receive purchase invoices.

FR-003

The system shall allow pharmacists to add multiple products within a single invoice.

FR-004

The system shall create batch records automatically from invoice items.

FR-005

The system shall increase inventory quantities upon invoice confirmation.

FR-006

The system shall detect existing batch numbers and update quantities instead of creating duplicate batch records.

Inventory Monitoring

FR-007

The system shall track inventory quantities for all products.

FR-008

The system shall update stock levels within 5 seconds after every inventory transaction.

FR-009

The system shall identify products whose available stock quantity is less than or equal to the reorder threshold.

Expiry Management

FR-010

The system shall track expiry dates at batch level.

FR-011

The system shall generate expiry alerts for batches expiring within 90 days.

FR-012

The system shall generate a Near Expiry Report.

FR-013

The Near Expiry Report shall display:

- * Product Name
- * Batch Number
- * Current Quantity
- * Expiry Date
- * Remaining Days to Expiry
- * Estimated Loss Value

FR-014

The system shall automatically identify expired batches.

FR-015

The system shall generate an Expired Products Report.

FR-016

The system shall allow pharmacists to dispose expired products.

FR-017

The system shall deduct disposed quantities from inventory automatically.

FR-018

The system shall calculate the financial loss caused by expired products.

FR-019

The system shall maintain disposal history for auditing purposes.

Reorder Recommendation

FR-020

The system shall generate reorder recommendations based on:

- * Last 6 months sales history
- * Current stock quantity
- * Reorder threshold
- * Product movement rate
- * Expiry risk

FR-021

The system shall display the recommended quantity with the calculation factors:

- * Average Monthly Sales
- * Current Stock
- * Movement Classification
- * Expiry Risk

FR-022

The pharmacist shall be able to approve the recommendation then reorder request status becomes Completed.

FR-023

The pharmacist shall be able to modify the recommendation.

FR-024

A justification shall be mandatory when modifying a recommendation.

FR-025

Modified recommendations shall require Branch Manager approval.

FR-026

The system shall showing Reorder Request Status:

- * Draft
- * Approved
- * Pending Manager Approval
- * Rejected
- * Completed

FR-027

A Quantity Number shall be mandatory when the Manager chooses Enter Custom Quantity.

Daily Replenishment Report

FR-028

The system shall generate a Daily Replenishment Report at the start of each business day.

FR-029

The report shall include:

- * Products requiring reorder
- * Products below reorder threshold
- * Pending reorder requests
- * Previously unresolved reorder items

FR-030

The report shall display:

- * Product Name
- * Current Stock
- * Reorder Threshold
- * Recommended Quantity
- * Reorder Status

FR-031

Products shall remain in the report until a new batch is received and inventory quantity exceeds reorder threshold.

FR-032

System shall import System shall import:

- * Transaction ID
- * Product ID
- * Quantity Sold
- * Sale Date

From an external POS system.

FR-033

Duplicate Transaction IDs shall be ignored.

FR-034

Imported transactions shall update inventory quantities.

Dashboards

FR-035

The system shall provide an Inventory Dashboard.

FR-036

The system shall provide a Sales Dashboard.

Audit Log

FR-037

The system shall record all reorder modifications and approvals:

- * User
- * Action
- * Old Value
- * New Value
- * Timestamp

FR-038

The system shall calculate the estimated financial loss of near-expiry products.

FR-039

The system shall allow users to view reorder request history.

• NON-FUNCTIONAL REQUIREMENTS

NFR-001 Performance

The system shall display any dashboard page within 3 seconds under normal operating conditions.

NFR-002 Performance

The system shall generate the Daily Replenishment Report within 30 seconds.

NFR-003 Performance

The system shall generate the Near Expiry Report within 30 seconds.

NFR-004 Availability

The system shall be available 99% of business hours excluding scheduled maintenance periods.

NFR-005 Security

Only authenticated users shall be allowed to access the system.

NFR-006 Security

Users shall only access functions permitted by their assigned role.

NFR-007 Security

All user Passwords shall be stored using secure one-way hashing algorithms.

NFR-008 Security

All critical inventory actions shall be recorded in the audit log.

NFR-009 Security

User sessions shall automatically expire after 15 minutes of inactivity.

NFR-010 Reliability

The system shall prevent duplicate Product IDs.

NFR-011 Reliability

The system shall prevent duplicate Batch Numbers for the same product.

NFR-012 Reliability

The system shall validate all mandatory fields before saving data.

NFR-013 Scalability

The system shall support at least 10 concurrent users without performance degradation.

NFR-014 Maintainability

System configuration values such as Reorder Thresholds and Expiry Alert Periods shall be configurable without code changes.

NFR-015 Usability

All reports shall support sorting and filtering.

NFR-016 Usability

Users shall be able to search products by Product ID or Product Name.

NFR-017 Usability

Users shall receive clear validation messages when invalid data is entered.

NFR-018 Auditability

Audit logs shall retain records for a minimum of 2 years.

NFR-019 Auditability

Each audit log record shall contain User, Action, Timestamp, Old Value, and New Value.

NFR-020 Backup

The system shall perform full daily database backup.

NFR-021 Recovery

The system shall restore system data within 4 hours after a critical failure.

NFR-022 Compatibility

The system shall Support latest two versions of Chrome, Edge and Firefox.

NFR-023 Localization

The system shall Support English and Arabic languages.

NFR-024 Data Integrity

Inventory quantities shall never become negative.

• User Stories

US-001

As a Pharmacist, I want to add new products to the system so that they can be tracked and managed efficiently.

US-002

As a Pharmacist, I want to add purchase invoices so that inventory can be updated automatically.

US-003

As a Pharmacist, I want to view expiry dates so that I can prioritize selling near-expiry products.

US-004

As a Pharmacist, I want to receive near expiry alerts so that I can reduce losses caused by expired medications.

US-005

As a Pharmacist, I want to review near-expiry products so that I can take actions to reduce expiry losses.

US-006

As a pharmacist, I want to review expired products so that I can avoid selling expired products.

US-007

As a Pharmacist, I want to dispose of expired products so that inventory remains accurate.

US-008

As a Pharmacist, I want reorder recommendations so that I can avoid stock shortages and overstocking.

US-009

As a Pharmacist, I want to review a daily replenishment report so that I can quickly identify products that require ordering.

US-010

As a Pharmacist, I want to monitor Inventory performance through dashboards so that I can make better decisions.

US-011

As a Branch Manager, I want to review modified reorder requests so that I can approve inventory decisions.

US-012

As a Branch Manager, I want to monitor sales performance through dashboards so that I can make better business decisions.

● Acceptance Criteria

AC-001

Add New Product

Scenario 1: Create Product Successfully

Given

- * Pharmacist is logged in.
- * Pharmacist has Product Management permission.

When

- * Pharmacist enters valid Product ID.
- * Enters Product Name.
- * Enters Category.
- * Enters Manufacturer.
- * Enters Supplier Name (free text).
- * Enters Reorder Threshold > 0.
- * Selects Product Status.
- * Clicks Save.

Then

- * Product record shall be created.
- * Product shall appear in Product List.
- * Success message shall be displayed.

Scenario 2: Duplicate Product ID

Given

- * Product ID already exists.

When

- * Pharmacist submits product form.

Then

- * Product shall not be saved.
 - * Error message “Product ID already exists” shall be displayed.
-

Scenario 3: Missing Mandatory Fields

Given

- * Pharmacist opens Add Product page.

When

- * Any mandatory field is empty.

Then

- * Product shall not be saved.
 - * Missing fields shall be highlighted.
 - * Validation message shall be displayed.
-

Scenario 4: Invalid Reorder Threshold

When

- * Pharmacist enters 0 or negative value.

Then

- * Product shall not be saved.
 - * Validation error shall appear.
-

AC-002

Add Purchase Invoice

Scenario 1: Create Invoice Successfully

Given

- * Pharmacist is logged in.
- * Product exists.

When

- * Pharmacist enters valid invoice data.
- * Adds invoice items.
- * Saves invoice.

Then

- * Invoice shall be created.
 - * Batch records shall be created.
 - * Inventory quantity shall be updated.
-

Scenario 2: Existing Batch Number

Given

- * Same Product ID.
- * Same Batch Number exists.

When

- * Invoice is saved.

Then

- * Existing batch quantity shall increase.
 - * No duplicate batch record shall be created.
-

Scenario 3: Expired Batch

When

- * Expiry Date is earlier than current date.

Then

- * Invoice item shall be rejected.
 - * Error message shall be displayed.
-

Scenario 4: Invalid Quantity

When

- * Quantity ≤ 0

Then

- * Item shall not be saved.
 - * Validation error shall be displayed.
-

Scenario 5: Near Expiry Batch

When

- * Batch expiry ≤ 90 days.

Then

- * Batch shall be saved.
 - * Batch status shall become Near Expiry.
 - * Batch shall appear in Near Expiry Report.
-

AC-003

Generate Near Expiry Alerts

Scenario 1: Alert Generation

Given

- * Active batch exists.
- * Remaining expiry days ≤ 90 .

When

- * Daily expiry job runs.

Then

- * Expiry alert shall be generated.
 - * Batch shall be marked Near Expiry.
 - * Batch shall appear in dashboard alerts.
-

Scenario 2: No Near Expiry Products

Given

- * No batches expire within 90 days.

When

- * Job runs.

Then

- * No alerts shall be generated.
-

AC-004

Near Expiry Report

Scenario 1: Display Report

Given

- * Near expiry batches exist.

When

- * Pharmacist opens report.

Then

- * System shall display:

- * Product Name
- * Batch Number
- * Quantity
- * Expiry Date
- * Remaining Days
- * Estimated Loss Value

Scenario 2: Empty Report

Given

- * No near expiry products exist.

When

- * Report is opened.

Then

- * "No records found" message shall appear.

AC-005

Generate Reorder Recommendation

Scenario 1: Recommendation Created

Given

- * Current Stock $\leq$ Threshold.
- * Sales history exists.

When

- * Reorder engine runs.

Then

- * System shall calculate recommendation using:
 - * Last 6 months sales
 - * Current Stock
 - * Threshold
 - * Movement Classification
 - * Expiry Risk

Scenario 2: Calculated Quantity $\leq$ 0

When

- * Calculated recommendation $\leq$ 0.

Then

- * Recommendation shall not be generated.
-

Scenario 3: Product Above Threshold

When

* Current Stock > Threshold.

Then

* Recommendation shall not be generated.

Scenario 4: Daily Report Integration

When

* Recommendation is generated.

Then

* Product shall be added automatically to Daily Replenishment Report.

Scenario 5

No Sales History

Given no sales history exists

Then

Recommendation shall not be generated

AC-006

Branch Manager Approval

Scenario 1: Approve Pharmacist Quantity

When

- * Manager selects Approve Pharmacist Quantity.

Then

- * Final Approved Quantity = Requested Quantity.
 - * Status = Completed.
-

Scenario 2: Approve Original Recommendation

When

- * Manager selects Original Recommendation.

Then

- * Final Approved Quantity = Recommended Quantity.
 - * Status = Completed.
-

Scenario 3: Enter Custom Quantity

When

- * Manager enters a custom quantity.

Then

- * Manager custom quantity requires justification.
 - * Final Approved Quantity = Manager Quantity .
 - * Status = Completed.
-

Scenario 4: Reject Request

When

- * Manager rejects request.

Then

- * Status = Rejected.

- * Audit log shall be created.

Scenario 5: Duplicate Approval

Given

- * Request already processed.

When

- * Manager attempts approval again.

Then

- * System shall prevent action.

- * Warning message shall appear.

AC-007

Daily Replenishment Report

Scenario 1: Daily Generation

When

- * Business day starts.

Then

* Report shall be generated automatically.

Scenario 2: Report Contents

Then

* Report shall contain:

- * New reorder items
 - * Pending reorder items
 - * Unresolved reorder items
-

Scenario 3: Product Persistence

Given

* Product still below threshold.

Then

* Product remains in report.

Scenario 4: Product Removal

Given

- * New batch received.
- * Stock > Threshold.

Then

* Product shall be removed from report.

Scenario 5: Approve Recommendation

Given

- * Recommendation exists.

When

- * Pharmacist clicks Approve.

Then

- * Status becomes Approved.
 - * Audit log shall be created.
-

Scenario 6: Modify Recommendation

When

- * Pharmacist changes quantity.

Then

- * Justification field becomes mandatory.
 - * Status becomes Pending Manager Approval.
-

Scenario 7: Missing Justification

When

- * Pharmacist submits modified quantity without justification.

Then

- * Submission shall be rejected.
- * Validation message shall appear.

AC-008

Inventory Dashboard

Scenario 1: Dashboard Display

When

* Pharmacist opens dashboard.

Then

- * Dashboard shall display:
 - * Reorder Products Count
 - * Near Expiry Count
 - * Expired Count
 - * Pending Approvals Count
 - * Fast Moving Products
 - * Slow Moving Products

AC-009

Sales Dashboard

Scenario 1: Dashboard Display

When

* Branch Manager opens dashboard.

Then

- * Dashboard shall display:
 - * Total Sales
 - * Inventory Value
 - * Sales Trend
 - * Estimated Expiry Loss

* Pending Approvals

AC-010

Expired Products Report

Scenario 1: Display Expired Products

Given

* Expired batches exist.

When

* Report is opened.

Then

* System shall display:

- * Product Name
- * Batch Number
- * Quantity
- * Expiry Date
- * Expired Days
- * Estimated Loss

Scenario 2: No Expired Products

When

* No expired products exist.

Then

* Message “No expired products found” shall be displayed.

AC-011

Dispose Expired Products

Scenario 1: Dispose Successfully

Given

- * Batch status = Expired.

When

- * Pharmacist enters:
 - * Disposal Date
 - * Disposal Reason
 - * Quantity

Then

- * Inventory quantity shall decrease.
- * Disposal transaction shall be recorded.
- * Audit log shall be created.
- * Loss value shall be updated.

Scenario 2: Quantity Exceeds Available

When

- * Disposed Quantity > Available Quantity.

Then

- * Disposal shall be rejected.
- * Validation error shall appear.

Scenario 3: Cancel Disposal

When

- * Pharmacist cancels process.

Then

- * No data shall be changed.

AC-012

Audit Log

Scenario 1: Reorder Modification

When

- * Pharmacist modifies recommendation.

Then

- * Audit Log shall store:

- * User
- * Action
- * Old Value
- * New Value
- * Timestamp

Scenario 2: Manager Approval

When

- * Manager approves/rejects request.

Then

- * Audit record shall be created.

AC-013
POS Integration

Scenario 1: Import Sales

Given

- * New sales transactions exist in POS.

When

- * Import process runs.

Then

- * Sales transactions shall be imported successfully.
- * Inventory quantities shall be updated.

Scenario 2: Duplicate Transaction

When

- * Same transaction is imported again.

Then

- * Duplicate transaction shall not be created.

- **Data Dictionary**

1- Product

Field	Type
Product ID	Number
Product Name	Text
Category	Text
Current Stock Quantity	Number
Manufacturer	Text
Unit Of Measure	Text
Supplier Name	Text
Reorder Threshold	Number
Product Status	Active / Inactive
Created Date	Date

2- Purchase Invoice

Field	Type
Invoice Number	Text
Supplier Name	Text
Invoice Date	Date
Total Cost	Decimal
Created By	User

3- Invoice Item

Field	Type
-------	------

Invoice Number	Text
Product Name	Text
Product ID	Number
Batch Number	Text
Quantity	Number
Purchase Price	Number
Selling Price	Number
Expiry date	Date

4- Batch

Field	Type
Batch Number	Text
Product ID	Number
Quantity	Number
Purchase Price	Decimal
Selling Price	Decimal
Expiry Date	Date
Invoice Number	Text
Received Date	Date
Batch Status	Active, Near Expiry, Expired, Disposed, Consumed

5- Reorder Request

Field	Type
Request ID	Number
Product ID	Number
Recommended Quantity	Number
Requested Quantity	Number
Justification	Text
Approval Status	Text
Created Date	Date
Final Approved Quantity	Number
Approved By	Text
Approval Date	Date

6- User

Field	Type
User ID	Text
User Name	Text
Email	Text
Password	Text
Role	Text
Status	Text

7- Inventory Transaction

Field	Type
Transaction ID	Number
Product ID	Number
Batch Number	Text
Quantity	Number
Transaction Type	Text
Transaction Date	Date

8- Daily Replenishment Report

Field	Type
Product Name	Text
Current Stock	Number
Threshold	Number
Recommended Quantity	Number
Status	Draft, Approved, Pending Manager Approval, Completed, Rejected
Action	Approve / Modify

• Use Case:

UC-01

Add New Product

Primary Actor

Pharmacist

Trigger

A new medication needs to be registered in the system.

Preconditions

- * User is logged in.
- * User has permission to add products.
- * Product does not already exist.

Main Flow

1. Pharmacist selects "Add Product".
2. System displays product creation form.
3. Pharmacist enters:
 - * Product ID
 - * Product Name
 - * Category
 - * Manufacturer
 - * Supplier Name
 - * Reorder Threshold
 - * Product Status
4. Pharmacist submits the form.
5. System validates the entered data.
6. System creates the product record.
7. System displays a success message.

Alternative Flows

AF-01

If Product ID already exists:

1. System displays error message.
2. Product creation is cancelled.

AF-02

If mandatory data is missing:

1. System highlights missing fields.
2. Product is not saved.

Exception Flows

EF-01

If database connection fails:

1. System displays failure message.
2. Product is not created.

Post Conditions

* Product is available in the system.

UC-02

Receive Purchase Invoice

Primary Actor

Pharmacist

Trigger

A new Invoice of an existing products is received.

Preconditions

- * User is logged in.
- * User has permission to add batches.
- * Products exist in the system.

Main Flow

1. Pharmacist clicks "Receive New Invoice".
2. System displays invoice form.
3. Pharmacist enters:
 - * Invoice Number
 - * Supplier Name
 - * Invoice Date
4. Pharmacist starts adding invoice items.
5. For each item:
 - * Search Product
 - * Enter Batch Number
 - * Enter Quantity
 - * Enter Purchase Price
 - * Enter Selling Price
 - * Enter Expiry Date
6. System validates item data.
7. Pharmacist saves invoice.
8. System store invoice and batch records.
9. System updates inventory.

Alternative Flows

AF-01

If Batch Number already exists for selected product:

1. System increases the batch quantity.
2. Batch is saved.

AF-02

If any item expiry date is within configured alert period:

1. item data is saved.
2. item batch automatically appears in Near Expiry Report.

Exception Flows

EF-01

If any item expiry date has already passed:

1. System rejects item adding.
2. Error message is displayed.

EF-02

If item quantity is less than or equal to zero:

1. System rejects item submission.
2. Error message is displayed.

Post Conditions

- * Batch is added successfully.
- * Inventory quantity is updated.

UC-03

Generate Near Expiry Alerts

Primary Actor

System

Trigger

Daily scheduled expiry check.

Preconditions

* Products with expiry dates exist.

Main Flow

1. System scans all active batches.
2. System identifies batches expiring within 90 days.
3. System generates expiry alerts.
4. System displays alerts on dashboard.
5. System includes products in Near Expiry Report.

Alternative Flow

- * No batches found within expiry period.
- * System completes process without generating alerts.

Exception Flow

- * Scheduled job fails.
- * Error logged in Audit Log.

Post Conditions

- * Near Expiry alerts become available.

UC-04

View Near Expiry Report

Primary Actor

Pharmacist

Trigger

Pharmacist opens Near Expiry Report.

Preconditions

- * User is logged in.

Main Flow

1. Pharmacist opens Near Expiry Report.
2. System retrieves batches expiring within 90 days.
3. System displays:
 - * Product Name
 - * Batch Number
 - * Quantity
 - * Expiry Date
 - * Remaining Days
 - * Estimated Loss Value
4. Pharmacist reviews report.

Alternative Flow

- * No near-expiry batches found.
- * System displays “No records found”.

Exception Flow

- * Report generation failure.
- * System displays error message.

Post Conditions

- * Near Expiry Report is displayed.

UC-05

Generate Reorder Recommendation

Primary Actor

System

Trigger

Daily Replenishment Job.

Preconditions

- * Product exists.
- * Sales history exists.

Main Flow

1. System detects low stock.
2. System analyzes:
 - * Reorder threshold
 - * Last 6 months sales
 - * Current stock
 - * Product movement
 - * Expiry risk
3. System calculates recommended quantity.
4. System creates reorder recommendation.
5. System adds product to Daily Replenishment Report.

Alternative Flows

AF-01

- * Product stock > Reorder Threshold.
- * No recommendation generated.

AF-02

- * Calculated quantity ≤ 0 .
- * No recommendation generated.

Exception Flows

EF-01

- * Inventory data unavailable.
- * Recommendation generation fails.

Post Conditions

* Reorder recommendation is available.

UC-06

View Daily Replenishment Report

Primary Actor

Pharmacist

Trigger

Beginning of business day.

Preconditions

* User is logged in.

Main Flow

1. Pharmacist opens Daily Replenishment Report.
2. System retrieves:
 - * New reorder items
 - * Pending reorder items
 - * Unresolved reorder items
3. System displays:
 - * Product Name
 - * Current Stock
 - * Reorder Threshold
 - * Recommended Quantity
 - * Status
4. Pharmacist either:
 - * Approves recommendation.
 - * Modifies recommendation.
5. System processes selected action.

Alternative Flows

AF-01

- * Pharmacist approves recommendation directly.
- * Status = Approved.

AF-02

- * Pharmacist modifies quantity.
- * Justification becomes mandatory.
- * Status = Pending Manager Approval.

AF-03

- * No pending reorder items.
- * System displays empty report.

Exception Flow

EF-01

- * Pharmacist submits modification without justification.
- * System rejects submission.

Post Conditions

- * Report is displayed.
- * Recommendation is approved or modified.

UC-07

Branch Manager Approval Workflow

Primary Actor

Branch Manager

Trigger

A modified reorder request is submitted.

Preconditions

- * Pharmacist modified recommendation.
- * Justification is provided.

Main Flow

1. Branch Manager opens pending approvals.
2. System displays:
 - * Recommended quantity
 - * Requested quantity
 - * Justification
3. Branch Manager chooses:
 - * Approve pharmacist quantity.
 - * Approve original quantity.
 - * Enter custom quantity.
 - * Reject
4. System records decision.

Alternative Flows

AF-01

- * Manager approves pharmacist quantity.
- * Status = Completed.

AF-02

- * Manager approves original recommendation.
- * Status = Completed.

AF-03

- * Manager enters custom quantity.
- * Justification is Mandatory.
- * Status = Completed.

AF-04

- * Manager rejects request completely.
- * Status = Rejected.

Exception Flow

EF-01

- * Request already processed.
- * System prevents duplicate approval.

Post Conditions

- * Final reorder quantity is approved.

UC-08

View Inventory Dashboard

Primary Actor

Pharmacist

Trigger

User opens dashboard.

Main Flow

1. Pharmacist opens Inventory Dashboard.
2. System displays:
 - * Products requiring reorder
 - * Near expiry products
 - * Pending approvals
 - * Already Expired products
 - * Fast-moving products
 - * Slow-moving products

Alternative Flow

- * No products require reorder.
- * Dashboard displays zero counts.

Exception Flow

- * Dashboard data unavailable.
- * Error displayed.

Post Conditions

- * Dashboard is displayed.

UC-09

View Sales Dashboard

Primary Actor

Branch Manager

Trigger

Manager opens Sales Dashboard.

Main Flow

1. Branch Manager opens dashboard.
2. System displays:
 - * Total sales
 - * Total inventory value
 - * Sales trends
 - * Estimated expiry loss
 - * Requests awaiting approvals
3. Manager reviews performance.

Alternative Flow

- * No sales data available.
- * Dashboard displays empty charts.

Exception Flow

- * Sales aggregation failure.
- * Error displayed.

Post Conditions

- * Sales insights are displayed.

UC-10

View Expired Products Report

Primary Actor

Pharmacist

Trigger

Pharmacist opens Expired Products Report.

Preconditions

- * User is logged in.

Main Flow

1. Pharmacist opens Expired Products Report.
2. System retrieves expired batches.
3. System displays:
 - * Product Name
 - * Batch Number
 - * Quantity
 - * Expiry Date
 - * Expired Days
 - * Estimated Loss
4. Pharmacist reviews report.

Alternative Flow

AF-01

If no expired products exist System displays:
No expired products found.

Post Conditions

* Expired Products Report displayed.

UC-11

Dispose Expired Products

Primary Actor

Pharmacist

Trigger

Pharmacist selects expired products for disposal.

Preconditions

- * User is logged in.
- * Expired batch exists.
- * Batch status = Expired.

Main Flow

1. Pharmacist opens Expired Products Report.
2. Pharmacist selects expired batch.
3. Pharmacist clicks Dispose.

4. System requests:
 - * Disposal Date
 - * Disposal Reason
 - * Quantity Disposed
5. Pharmacist enters required information.
6. Pharmacist confirms disposal.
7. System deducts quantity from inventory.
8. System updates batch status to Disposed.
9. System records financial loss.
10. System creates audit log.

Alternative Flow

AF-01

Disposed quantity exceeds available quantity.

1. System displays validation error.
2. Disposal request is rejected.

AF-02

User cancels disposal process.

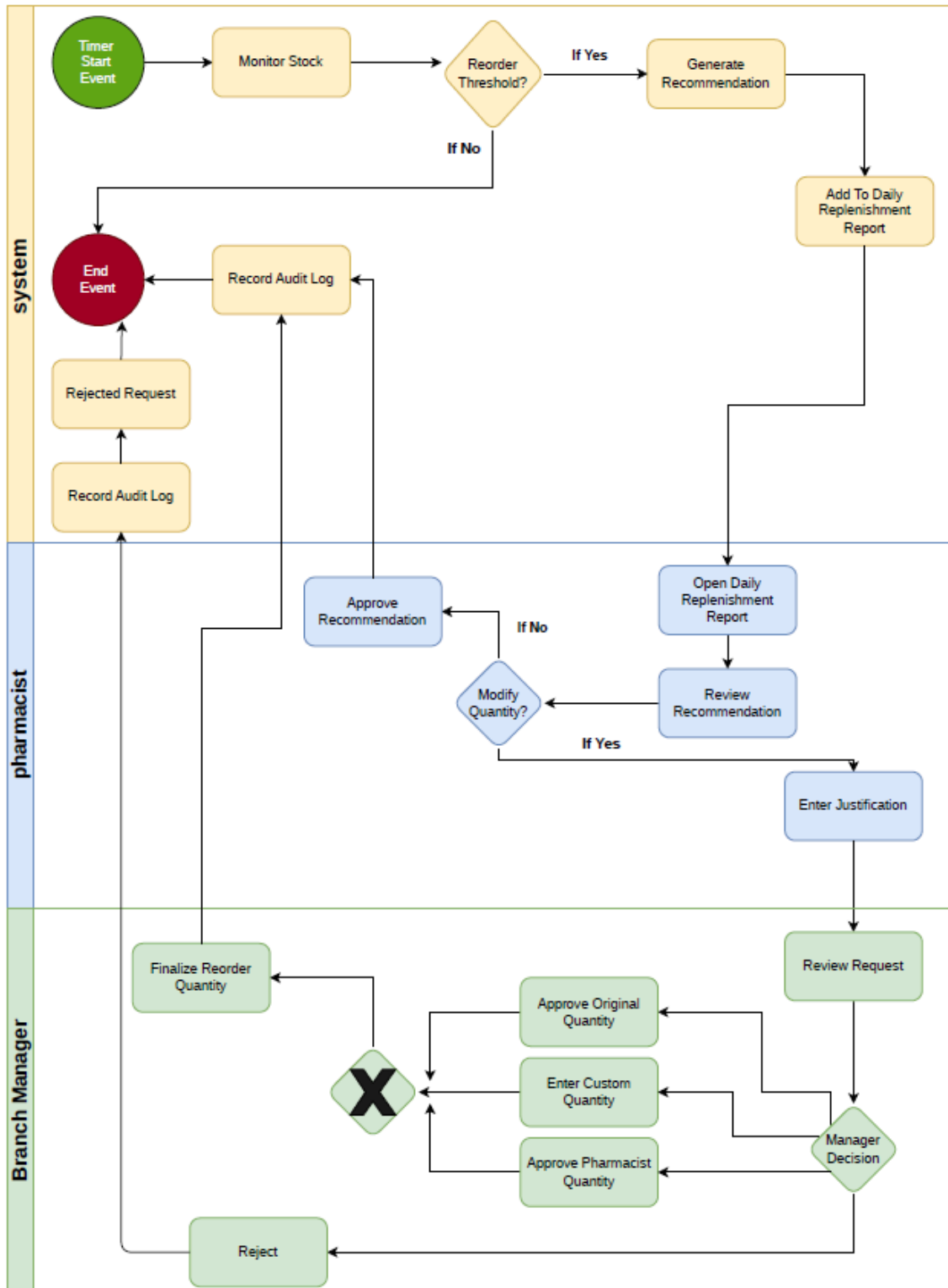
1. System discards changes.
2. Batch remains unchanged.

Post Conditions

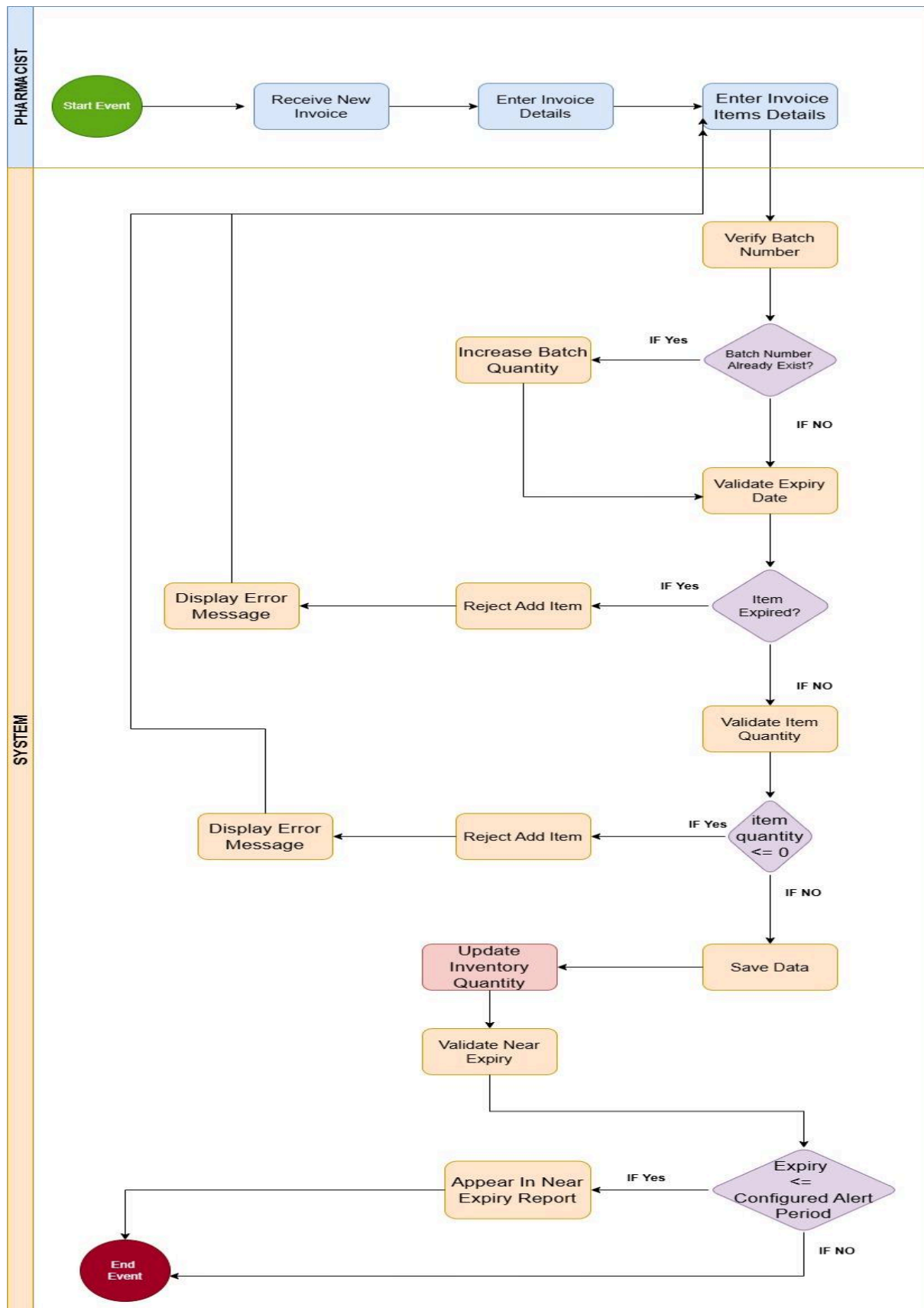
- * Inventory updated.
 - * Disposal transaction recorded.
 - * Loss value updated.
-

- BPMN

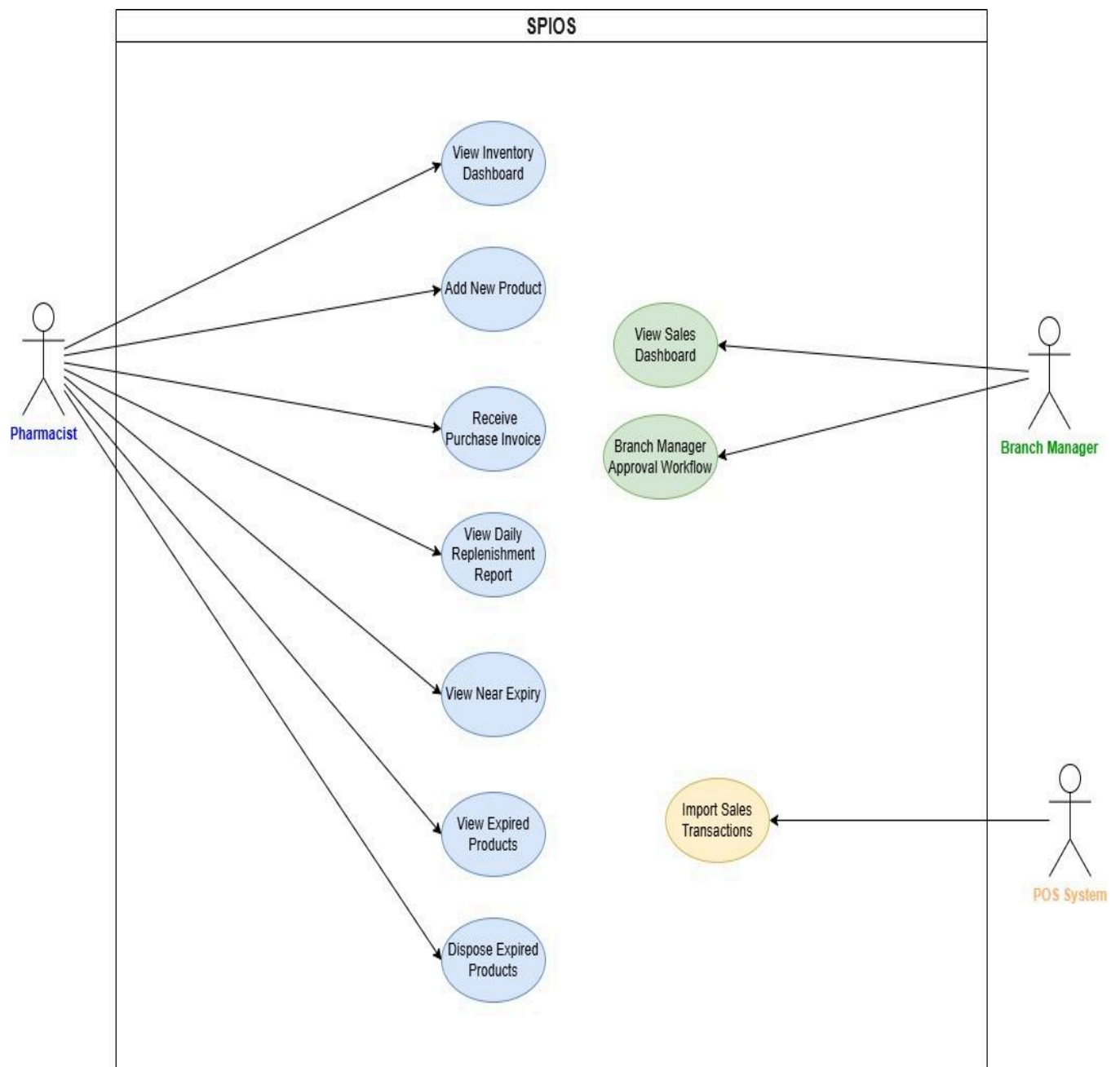
1- Inventory Reorder Process



1- Receive Purchase Invoice

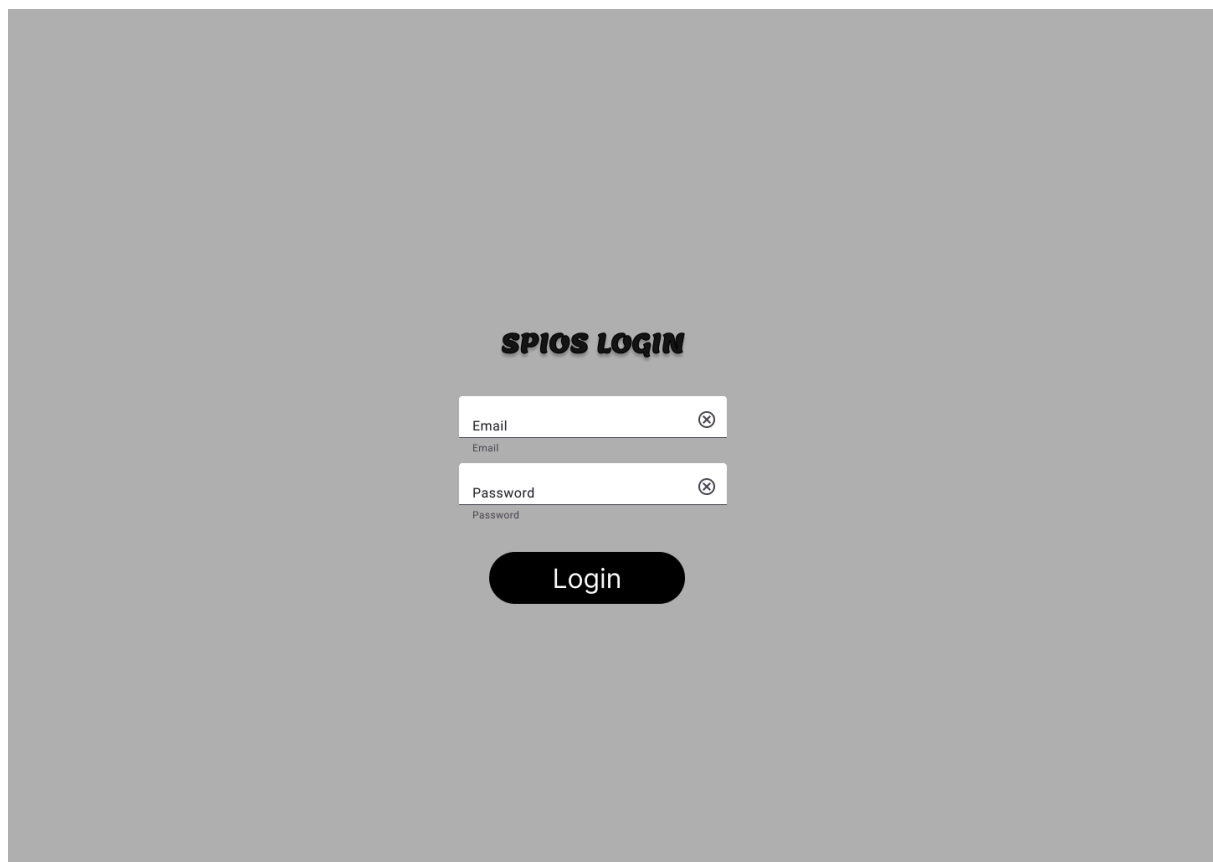


- Use Case Diagram



- **SPIOS Wireframes**

1- Login Page



A wireframe of a login page with a gray background. The title "SPIOS LOGIN" is centered in bold black text. Below it are two white input fields. The first field is labeled "Email" and has a small "x" icon in the top right corner. The second field is labeled "Password" and also has a small "x" icon in the top right corner. Below the input fields is a black rounded rectangle button with the word "Login" in white text.

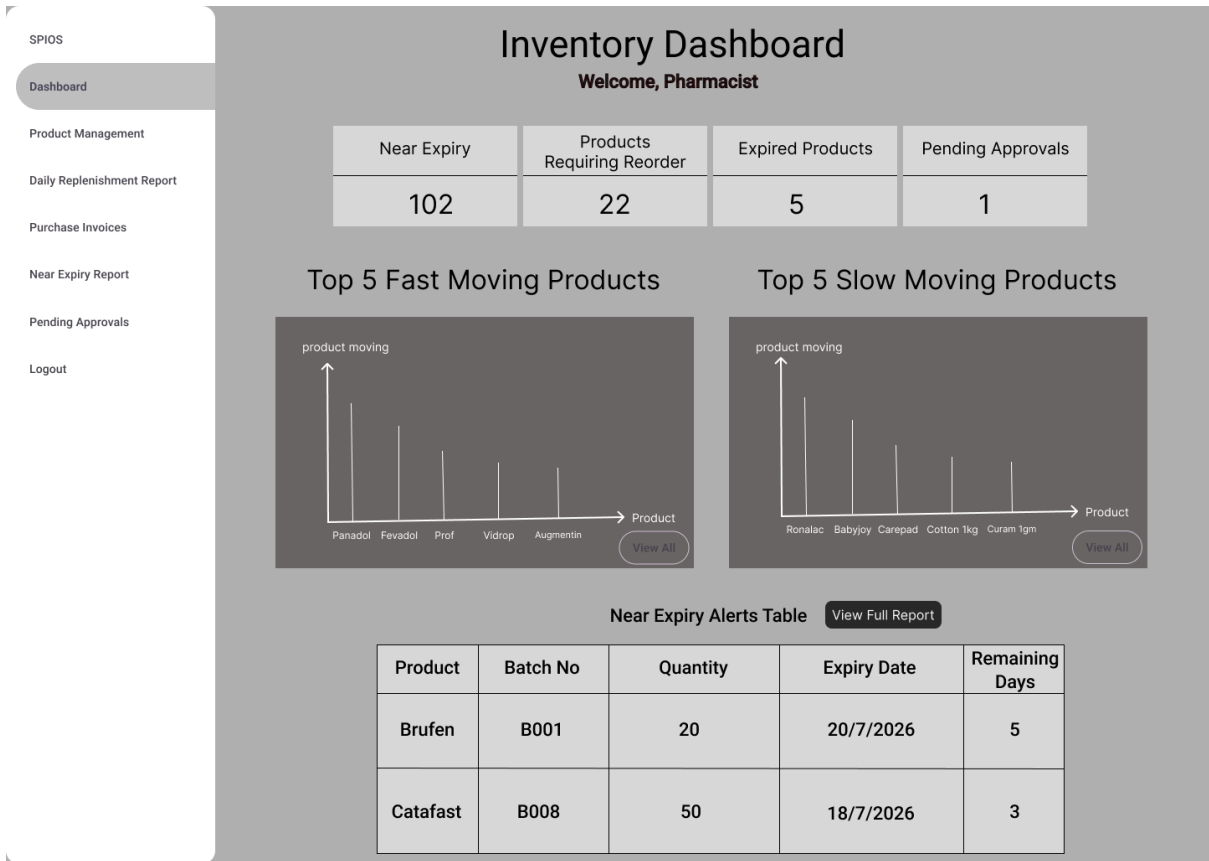
SPIOS LOGIN

Email

Password

Login

2- Inventory Dashboard



3- Product Management

SPIOS

Dashboard

Product Management

Daily Replenishment Report

Purchase Invoices

Near Expiry Report

Pending Approvals

Logout

Product Management

Section 1

Product ID	Product Name	Manufacturer	Supplier Name	Category	Unit Of Measure	Threshold	Stock Quantity	Status	Action
454612	Panadol	Ex	Sanabel	Medicine	Packet	40	150	Active	Edit
54646516	Fevadol	Ex	Sanabel	Medicine	Packet	20	100	Active	Edit
6546548	Augmentin	Ex	Sanabel	Medicine	Packet	1	9	Inactive	Edit
45685498	Feroglobin	Ex	Sanabel	Medicine	Packet	2	4	Inactive	Edit

Add Products

+

4- Daily Replenishment Report

SPIOS

Dashboard

Product Management

Daily Replenishment Report

Purchase Invoices

Near Expiry Report

Pending Approvals

Logout

Daily Replenishment Report

Section 1

Product Name	Current Stock	Threshold	Recommended Quantity	Status	Action
Panadol	35	40	165	Approved	Approve Modify
Fevadol	10	20	90	Pending Approval	Approve Modify
Augmentin	0	1	2	Completed	Approve Modify
Feroglobin	2	2	2	Completed	Approve Modify

5- Recommendation Modification

SPOS

Dashboard

Product Management

Daily Replenishment Report

Purchase Invoices

Near Expiry Report

Pending Approvals

Logout

Daily Replenishment Report

Section 1

Product Name	Current Stock				Action
Panadol	35				Approve Modify
Fevadol	10				Approve Modify
Augmentin	0				Approve Modify
Feroglobin	2				Approve Modify

Modify Recommendation?

Number

Requested Quantity

Text

Justification

Submit

Cancel

6- Purchase Invoices

SPIOS

Dashboard

Product Management

Daily Replenishment Report

Purchase Invoices

Near Expiry Report

Pending Approvals

Logout

Purchase Invoices

Add New Invoice

Section 1

Invoice Number	Supplier Name	Invoice Date	Total Cost	Created By	Action
SN9595944	Sanabel	23/6/2026	2500	Pharnacist	View Invoice
SN9595943	Sanabel	15/6/2026	1200	Pharmacist	View Invoice
SN9595942	Sanabel	10/6/2026	900	Branch manager	View Invoice
SN9595941	Sanabel	5/6/2026	1600	Pharmacist	View Invoice

7- Add New Invoice

SPIOS

Dashboard

Product Management

Daily Replenishment Report

Purchase Invoices

Near Expiry Report

Pending Approvals

Logout

Add New Invoice

Section 1

Invoice Number	SN9595945					
Product ID	Product Name	Batch Number	Quantity	Purchase Price	Selling Price	Expiry date
59554161	Panadol	B025	165	400	600	28/8/2028
321548	Fevadol	B680	90	150	350	23/5/2029
35216491	Augmentin	B078	2	100	150	14/1/2027
75415248	Feroglobin	B580	2	30	42	5/7/2027
⊕						

Save Invoice

8- Near Expiry Report

SPIOS

Dashboard

Product Management

Daily Replenishment Report

Purchase Invoices

Near Expiry Report

Pending Approvals

Logout

Near Expiry Report

Q Search

Export PDF

Section 1

Filters

Product Name	Batch Number	Quantity	Expiry Date	Remaining Days	Estimated Loss Value
Brufen	B001	20	20/7/2026	5	400
Catafast	B008	50	18/7/2026	3	1000
Adol	B055	60	15/8/2026	30	650
Prila	B002	10	20/9/2026	65	545

9- Manager Requests Approval

SPIOS

Dashboard

Branch Management

Requests awaiting approvals

Total inventory

Logout

Manager Requests Approval

Section 1

Product Name	Recommended Quantity	Requested Quantity	Justification	Action
Panadol	35	40	Becuase There is some one need 5 more Packets	Approve Requested Qty Approve Recommended Qty Custom Qty Reject
Fevadol	10	5	Becuase it's Price rising	Approve Requested Qty Approve Recommended Qty Custom Qty Reject
Augmentin	0	1	Reason	Approve Requested Qty Approve Recommended Qty Custom Qty Reject
Feroglobin	7	4	Reason	Approve Requested Qty Approve Recommended Qty Custom Qty Reject

10- Manager Quantity Customization

